



GA Groton Resource Optimization Workload Management and Analysis Tool (GROWMAT) External Presentation

Samuel Zhang

Herbie Bingham

GMP Analytics

Pfizer Sandwich x Pfizer Groton

Sandwich, UK

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Business Challenges

Workload Modelling and Resource Management

1. Uncertainties in workload distribution and schedule after business restructuring.
2. Legacy solution to workload management and scheduling does not scale well, isn't particularly robust, and takes a significant time to compute.
3. There lacks granularity in the legacy report, team leaders don't have the ability to check individual jobs that affect the total workload.



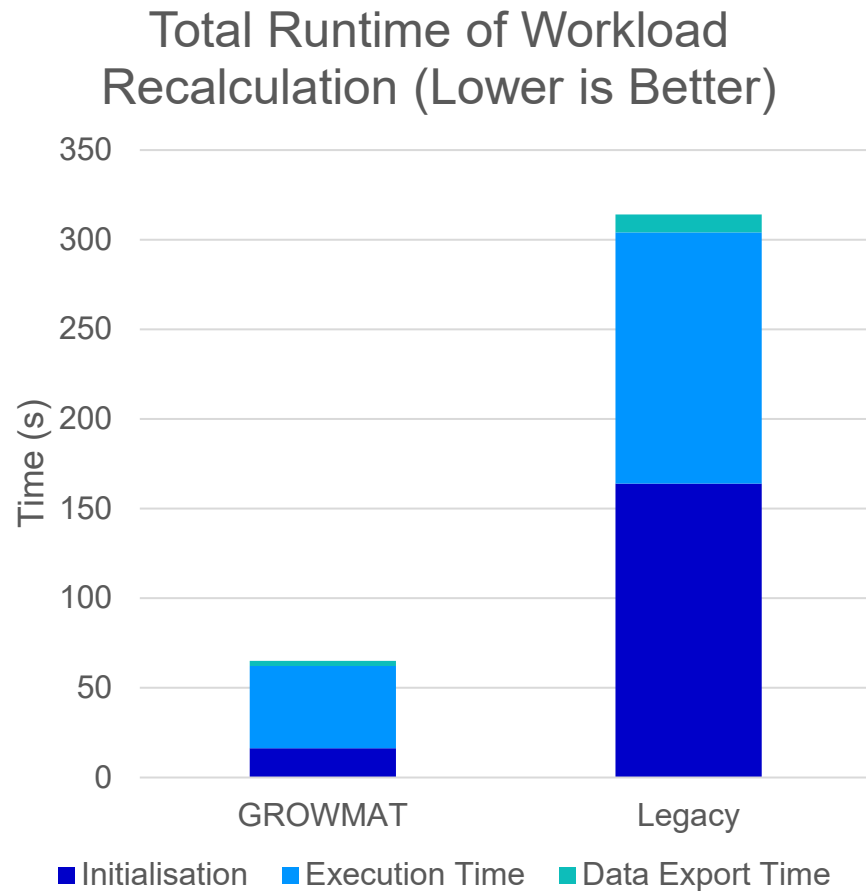
Business Solution

Workload Modelling and Resource Management

1. GROWMAT is designed to streamline Good Manufacturing Practice Analytics (GA)'s complex resource and workload management. It is set to replace Pfizer's legacy Excel-based GA Capacity Tool with improved features like advanced visualization through Spotfire, a dedicated website for scheduling, a more reliable codebase, and a support team for ongoing development.
2. Workload data are gathered from a variety of databases ranging from stability testing data from [LIMS](#); release, post packaging, excipients, and manufacturing data from [Clementine](#); to manually added jobs and rolling jobs. A unit of predicted workload [FTEs/day](#) (Full Time Employee per Day) is calculated for each job from historical data and automatically assigned an expected due date. Through different tabs, managers and team leaders can view not only a general workload distribution among teams, but they also have the granularity of viewing specific jobs affecting workload across time.
3. GROWMAT has two user facing components, a custom website for workload management (input) and a Spotfire view for data visualisation (output).

GROWMAT Features

Overview of benefits that GROWMAT has over the legacy tool (Groton Capacity Tool)



GROWMAT

- ☐ Responsive, simple, and user-friendly interface (Spotfire + Webpage)
- ☐ Much higher granularity in job specific data visualisation
- ☐ Bihourly automatic Spotfire visualisation update
- ☐ Bihourly automatic workload recalculation
- ☐ Concurrency proof workload scheduling
- ☐ 180 Days of database rollback, 3 tiers of disaster recovery plan
- ☐ Database aggregation via permanent service account
- ☐ Long-term support via Adrian's Team and/or DataSci

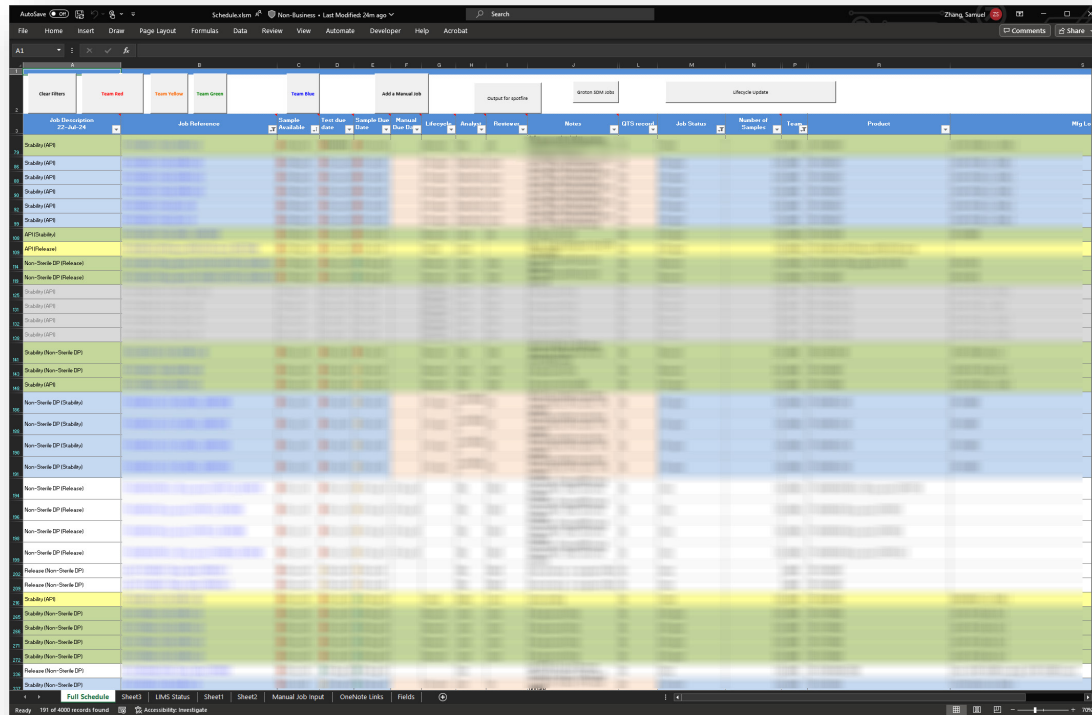
Legacy

- ☐ Clunky, error prone, complicated multi-workbook-dependent Excel interface
- ☐ Lacks graduality in job specific data visualisation
- ☐ Manual recalculation from analysts required
- ☐ Weekly manual workload recalculation from analysts needed
- ☐ Constant conflicts with concurrency issues, resulting in downtime
- ☐ Data rollback dependent on SharePoint only
- ☐ Database aggregation depends on user account
- ☐ No long-term support

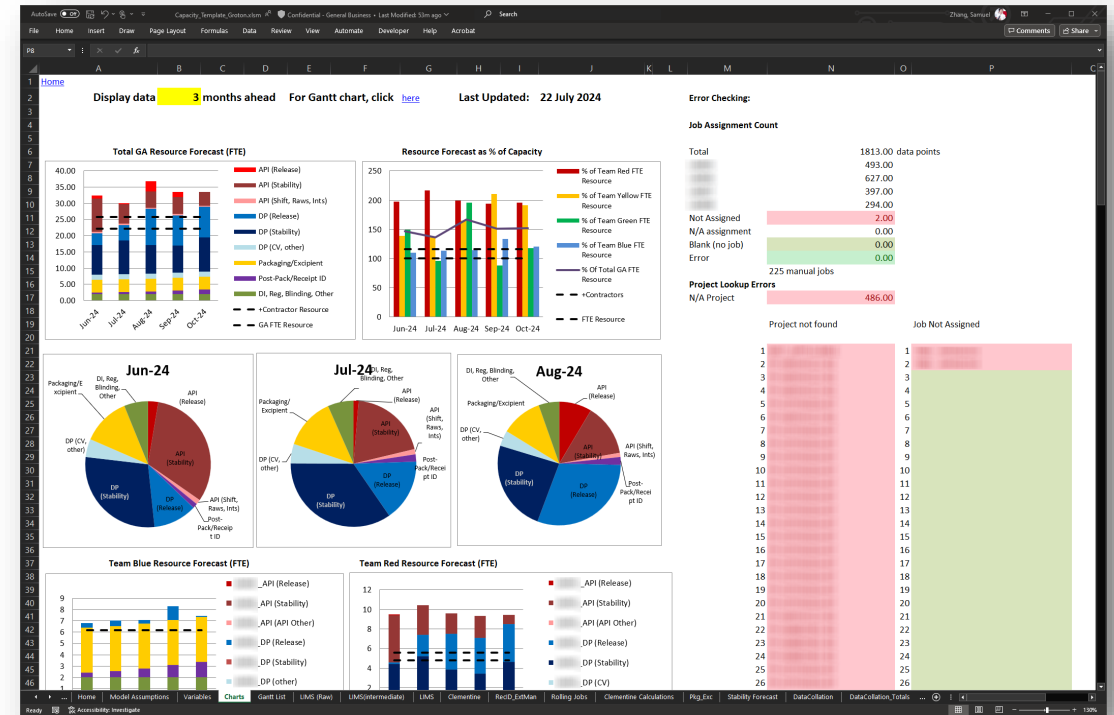
User Interface (Legacy)

The Legacy Excel based Workload Management Capacity Tool's Interface

Legacy Excel Workload Manager



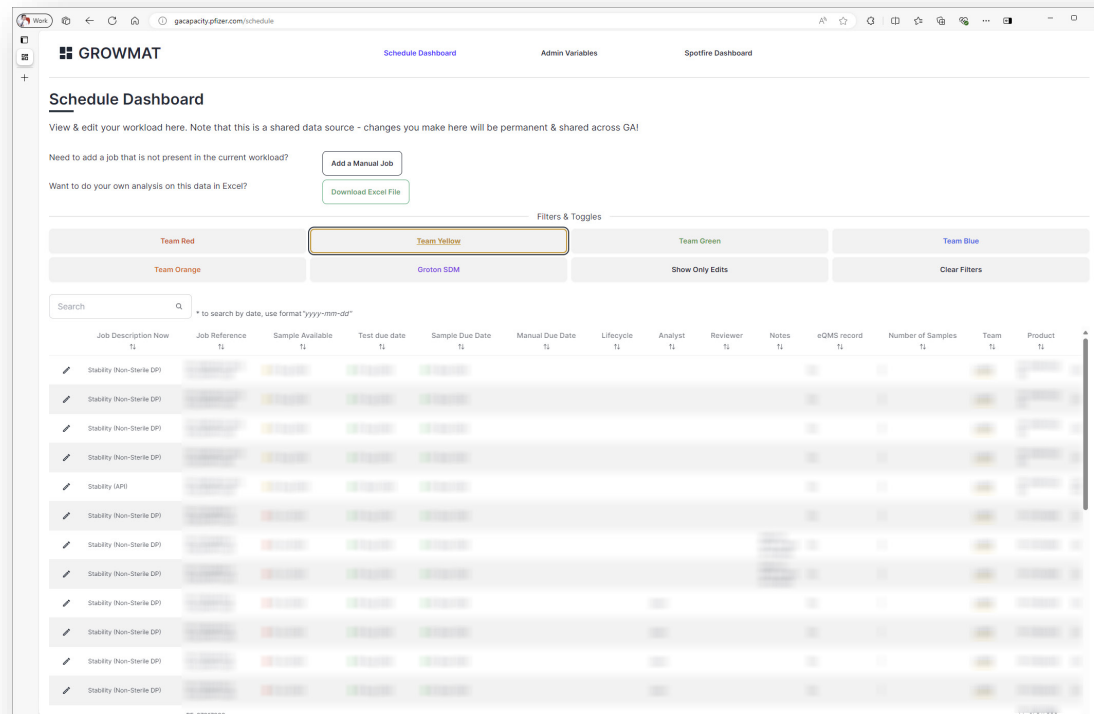
Legacy Excel Workload Visualiser



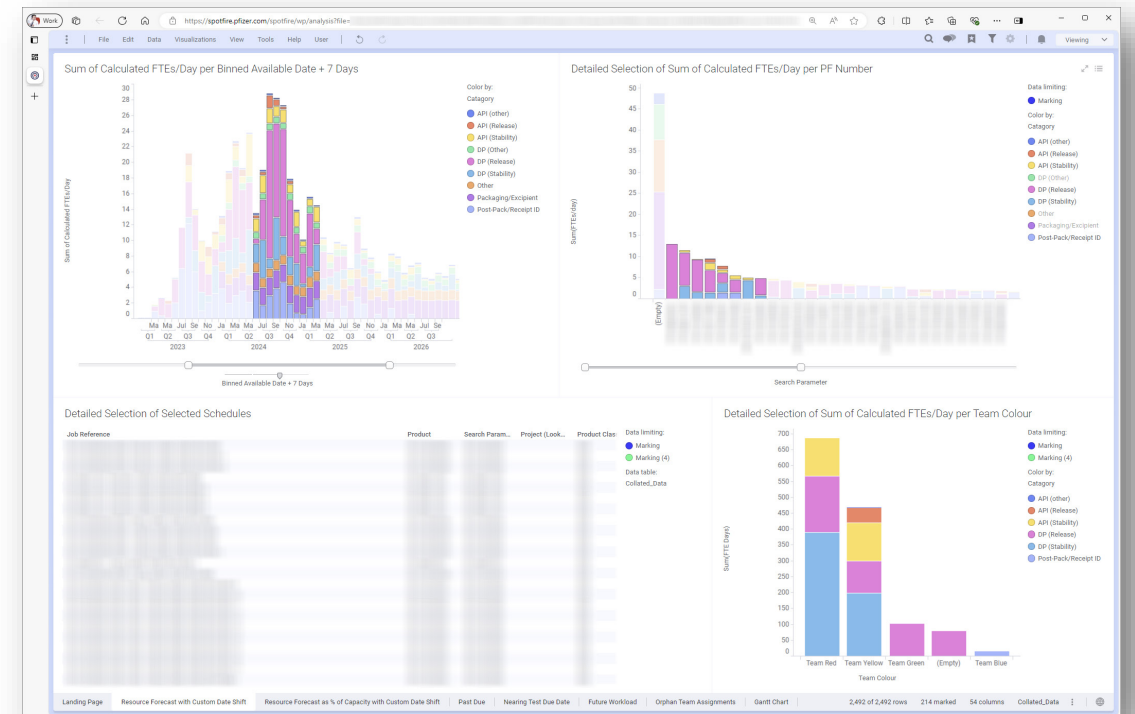
User Interface (GROWMAT)

GROWMAT's Web Based Interface

GROWMAT Web Workload Manager



GROWMAT Spotfire Workload Visualiser



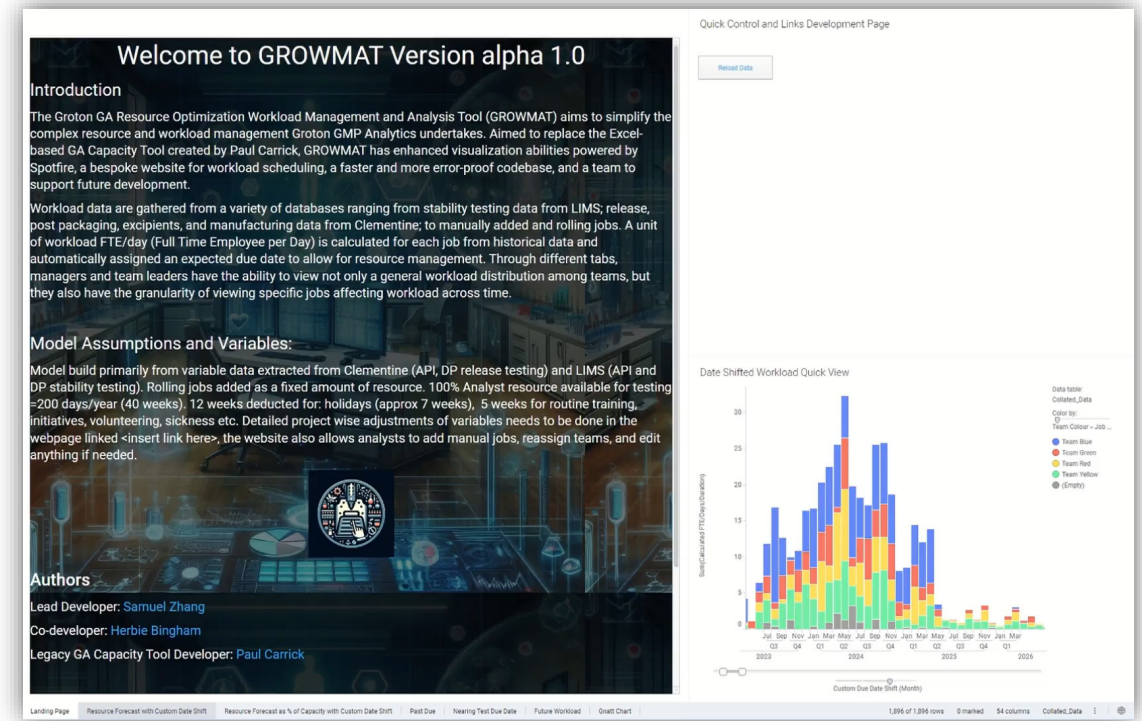
GROWMAT Spotfire Visualisation 1

Identifying Scheduled Jobs via FTE/Days

GROWMAT

- Spotfire's interactive approach allows the user to visualize any specific views independent from affecting source data and other users
- Much improved data aggregation allows a much more granular database where users can find any specific jobs within seconds through selecting visualisations
- Web and cloud-based approach simplifies user experience, allowing analysts to focus more on meaning behind the data rather than bug fixing and running separate visualisations locally.

Demonstration



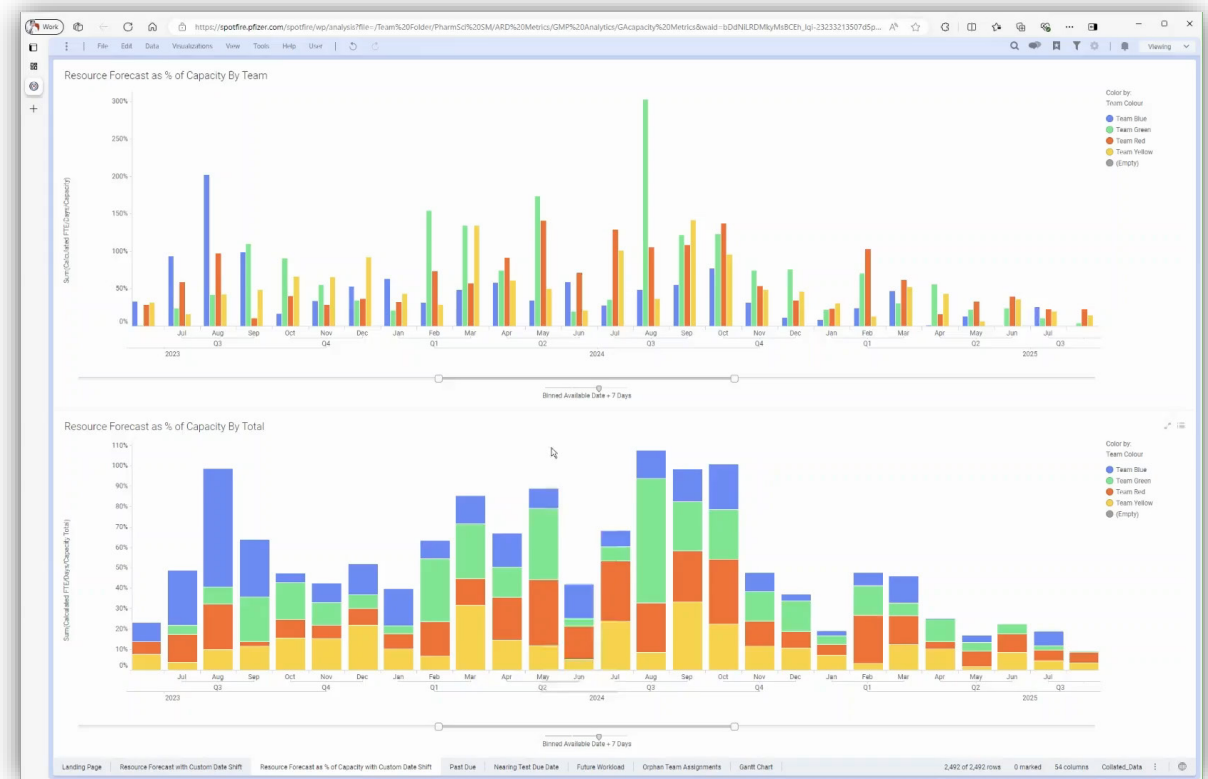
GROWMAT Spotfire Visualisation 2

Variety of Workload related Visualisations

GROWMAT

- Resource Forecast as % of Capacity
 - Allowing team leads and managers to easily identify if the current workload exceeds or is behind the target workload calculated from the current FTE working hours.
- Past Due
 - Quickly identifying any jobs that are at least 30 days past due date and isn't marked as completed
- Nearing Testing Due Date
 - Flags any jobs due within 7 days that is yet to be marked as completed
- Future Workload
 - Tracks any jobs due in the next month

Demonstration



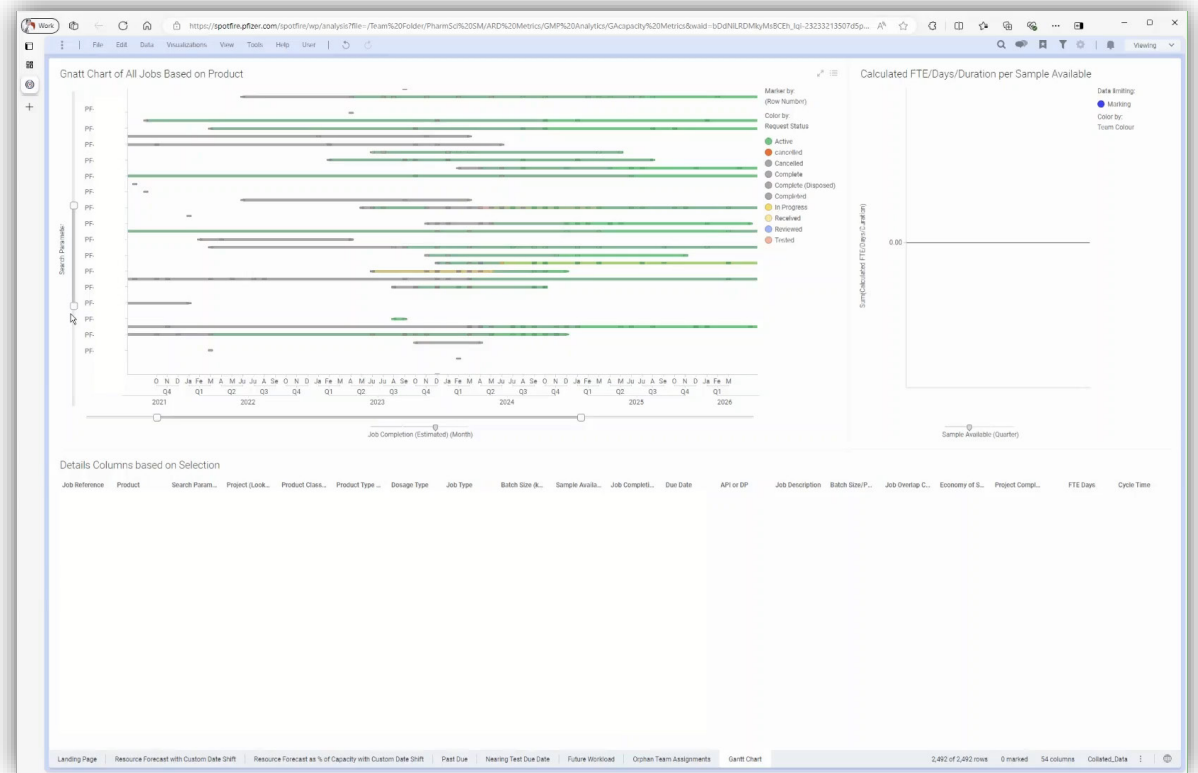
GROWMAT Spotfire Visualisation 3

Gantt Chart of all Groton GA Products

GROWMAT

- An overview of all projects (completed or cancelled) currently in GA Groton's portfolio. With time being the horizontal axis, each dots on the project indicates one assignment/task triggered by a due date.

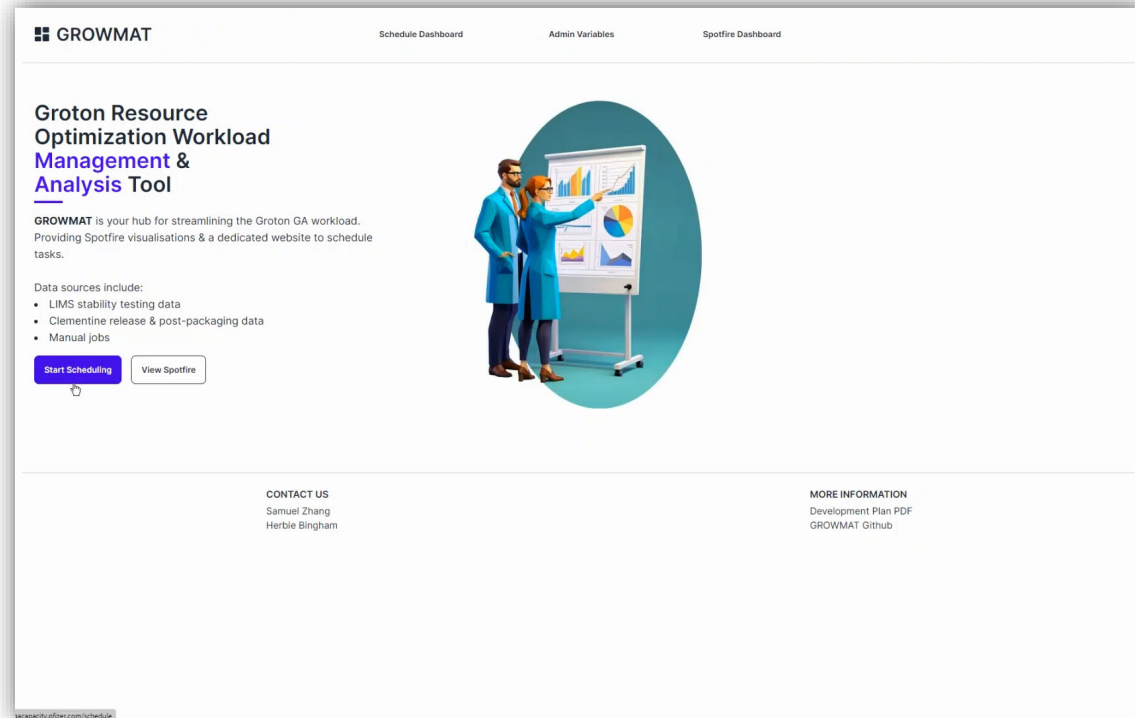
Demonstration



GROWMAT Workload Scheduler 1

Search, Filter, and Edit Job

GROWMAT



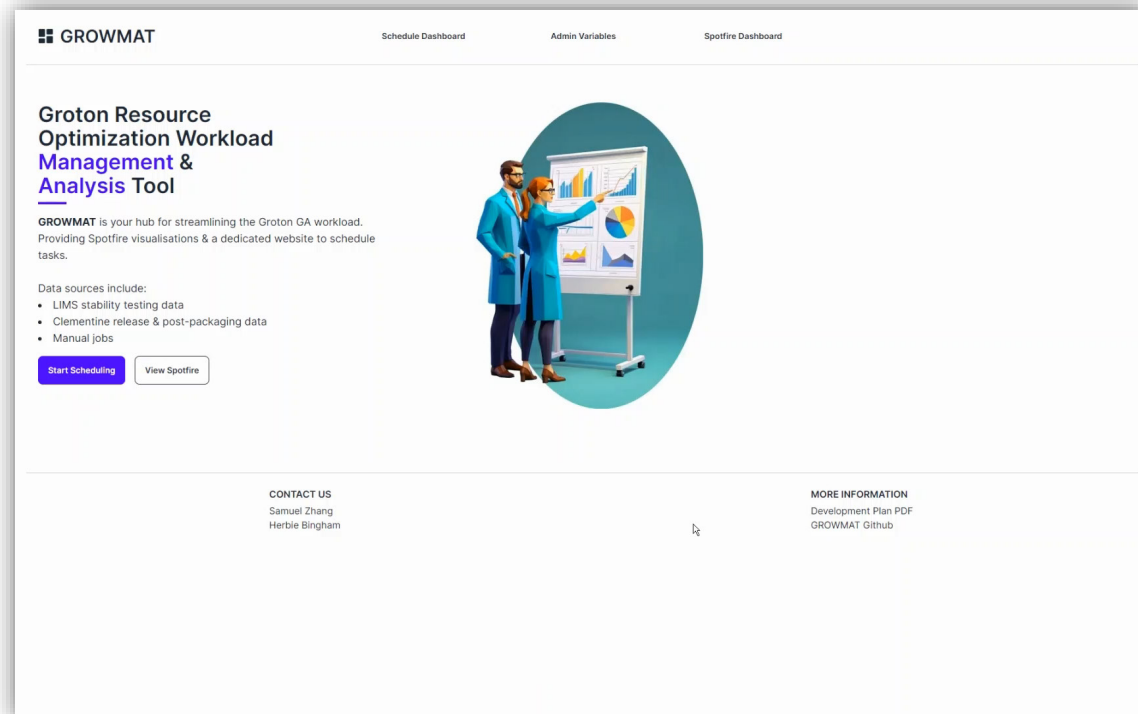
GROWMAT

- GROWMAT Webapp Allows the user to quickly filter all scheduled jobs, search through them, and easily edit any job status, due dates or team assignment with a few clicks.
- Since all the calculations are done in the cloud, the user will not experience any slowdown unlike the Legacy tool.

GROWMAT Workload Scheduler 2

Adding Manual Jobs

GROWMAT



GROWMAT

- An easy way to add any additional manual jobs.
- The restrictive nature of the webapp and forced field validation is for data integrity. Only allowing users to change one field at a time and disallowing wildcard characters prevents users breaking the backend calculation.
- Standard user is unable to delete entries, further enforcing data integrity.

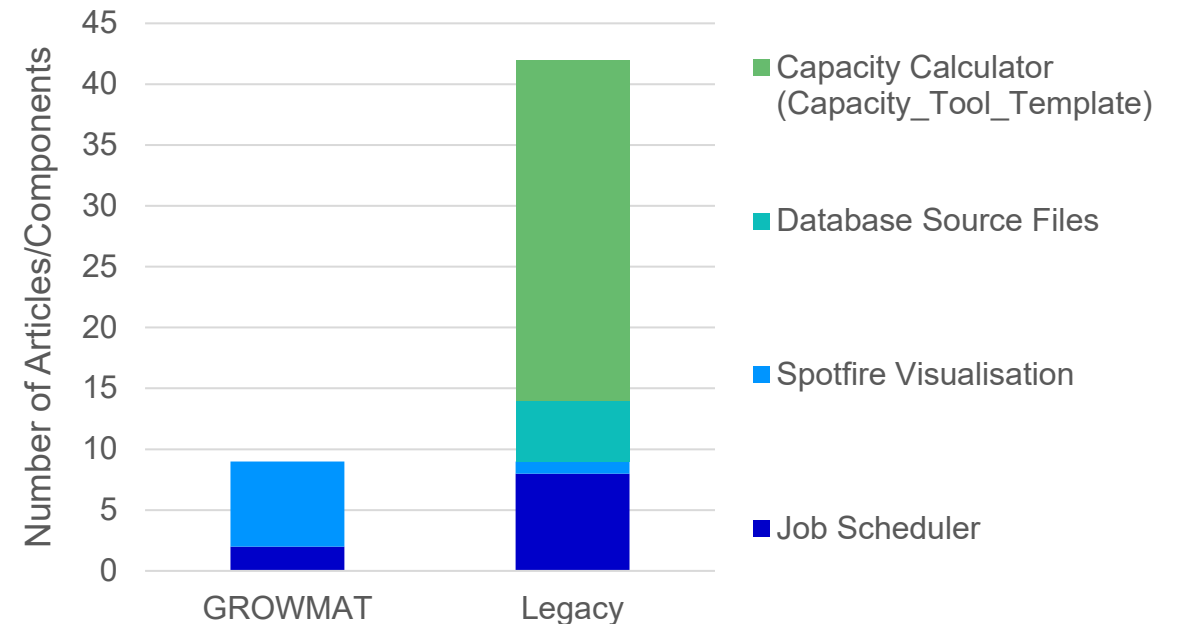
User Experience

Highlighting the Benefits of Reducing User Exposed Components

Benefits of Reducing Number of Articles

- Complex projects have the inherit issue with more points of failure. To minimise the issue, most of GROWMAT's components are not exposed to the end user, reducing the risk of user breaking the system.
- All GROWMAT's user facing components are web based, allowing user to manage, visualise, and edit workload completely through the web.
- The backend of GROWMAT is cloud based and hidden from the user. Any specific requests such as recovering data, adding new features, deleting records should be done via request only to preserve data integrity and system stability.

User Exposed Components (Lower is Better)

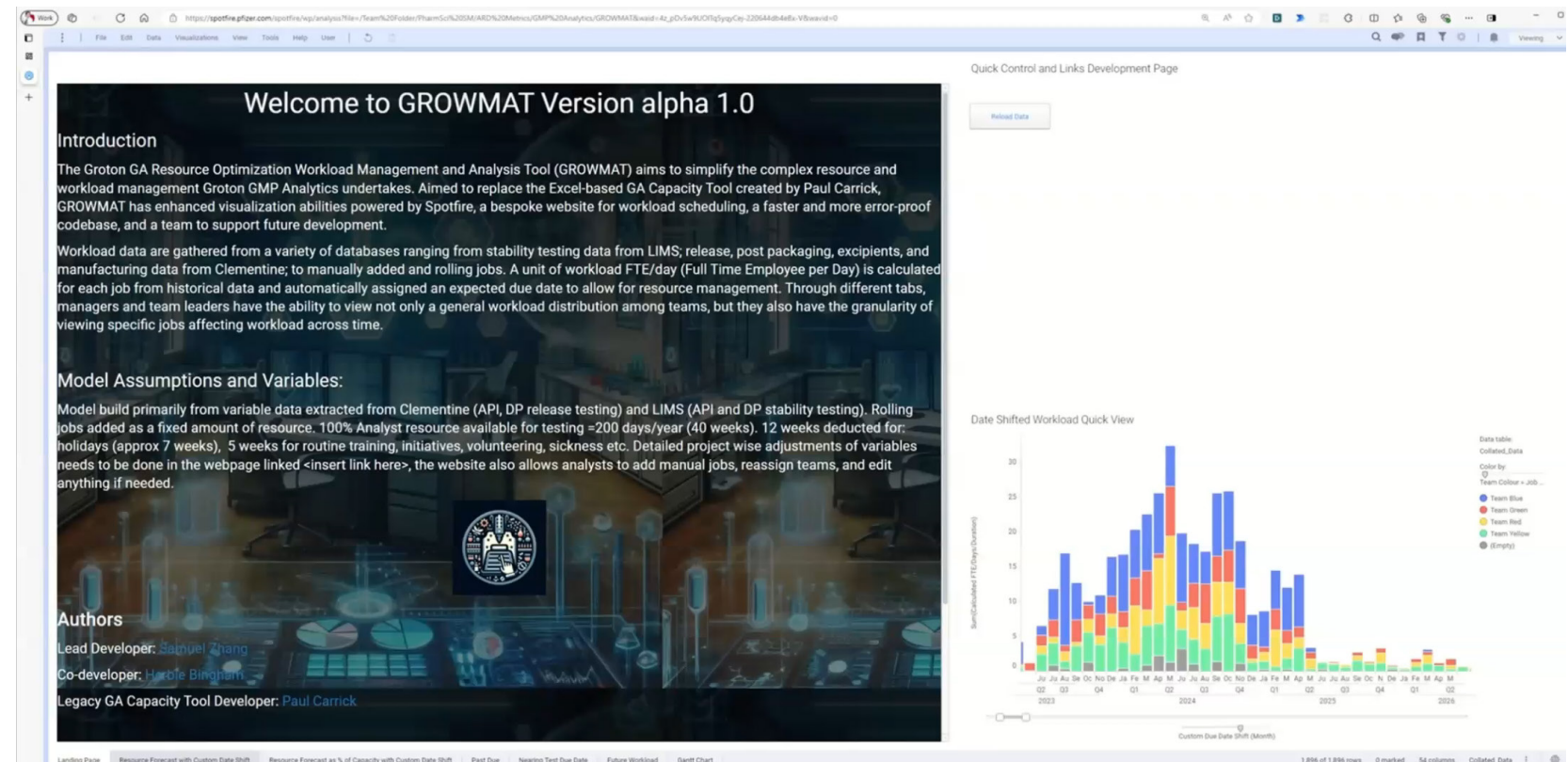


GROWMAT Use Case Example 1

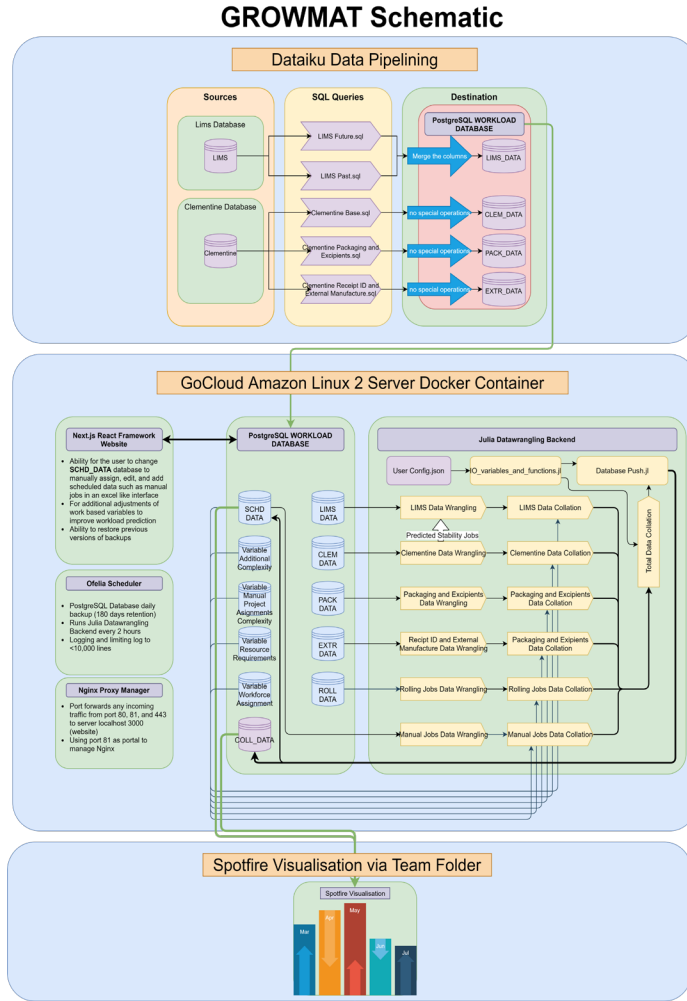
GROWMAT

- A GA Team Leader felt that there are overwhelming work this month (Sep 2024) allocated to their team. They want to investigate what jobs are causing the issue and would like to reassign the task to another team.

Demonstration



Technical Details on GROWMAT's Backend



Components of GROWMAT

- GROWMAT has 7 interlinked components functioning together, the data source comes from a **Dataiku Data Pipelining** schedule, the data is then pushed to a **GoCloud Amazon Linux 2 Server Docker Container** with 5 components that performs data storage (**PostgreSQL**), data wrangling (**Julia Backend**), task scheduling (**Ofelia**), and web server hosting (**Next.js on React Framework** and **Nginx Proxy Manager**). The output of which is then visualised using **Spotfire**.



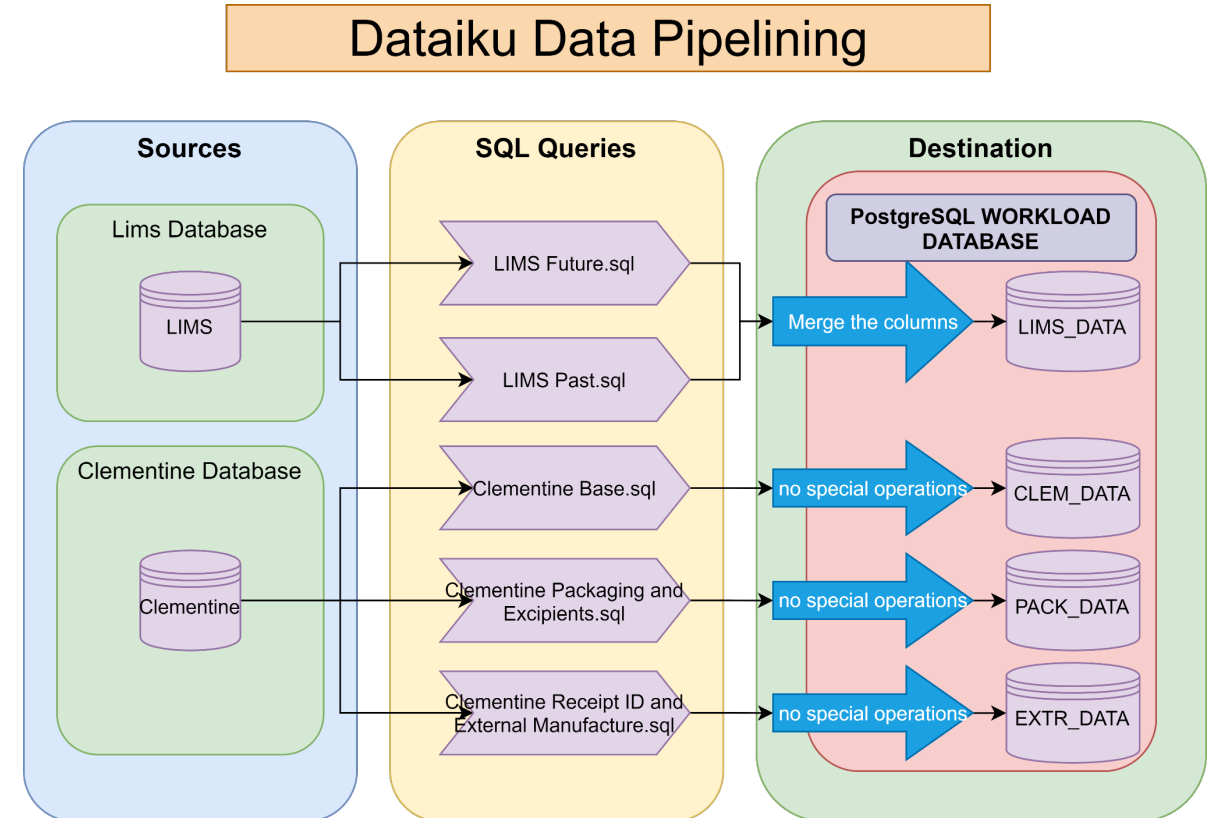
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Component 1: Dataiku Data Acquisition

Obtaining Data from Source Database Independent from GoCloud Server

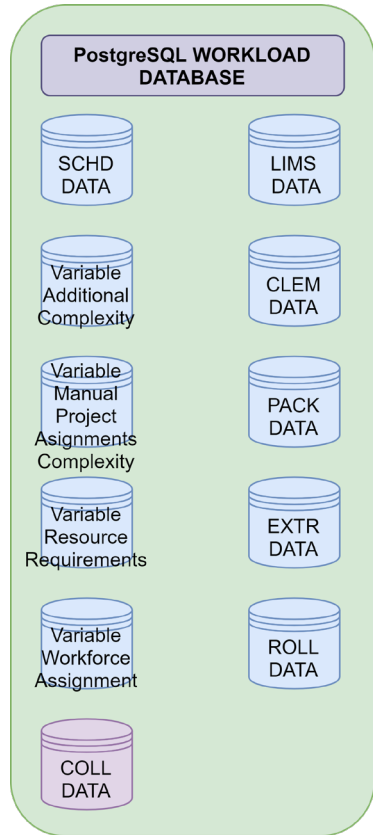
Dataiku Pipelining

- Current and future project stability data are queried from the LIMS database, aggregated, and put into GROWMAT's own PostgreSQL database by a Dataiku Data Pipelining routine that runs daily.
- Project release data, packaging and excipients data, receipt ID and external manufacturing data are also put into GROWMAT with the same Dataiku routine.



Component 2: PostgreSQL

GROWMAT: Storing Databases Using Docker Containerised PostgreSQL



PostgreSQL Database

- GROWMAT lives under the GoCloud Amazon Linux 2 server, within GROWMAT there are a few more components working together to make the entire service function. All components are containerised and deployed using Docker containers allowing easy management, deployment, and redeployment to other servers if needed.
- One of these containers acts as the data storage, it uses industry standard PostgreSQL to store 11 tables.
- Every column has data type, collation defined for data integrity. Every table has a unique key column.

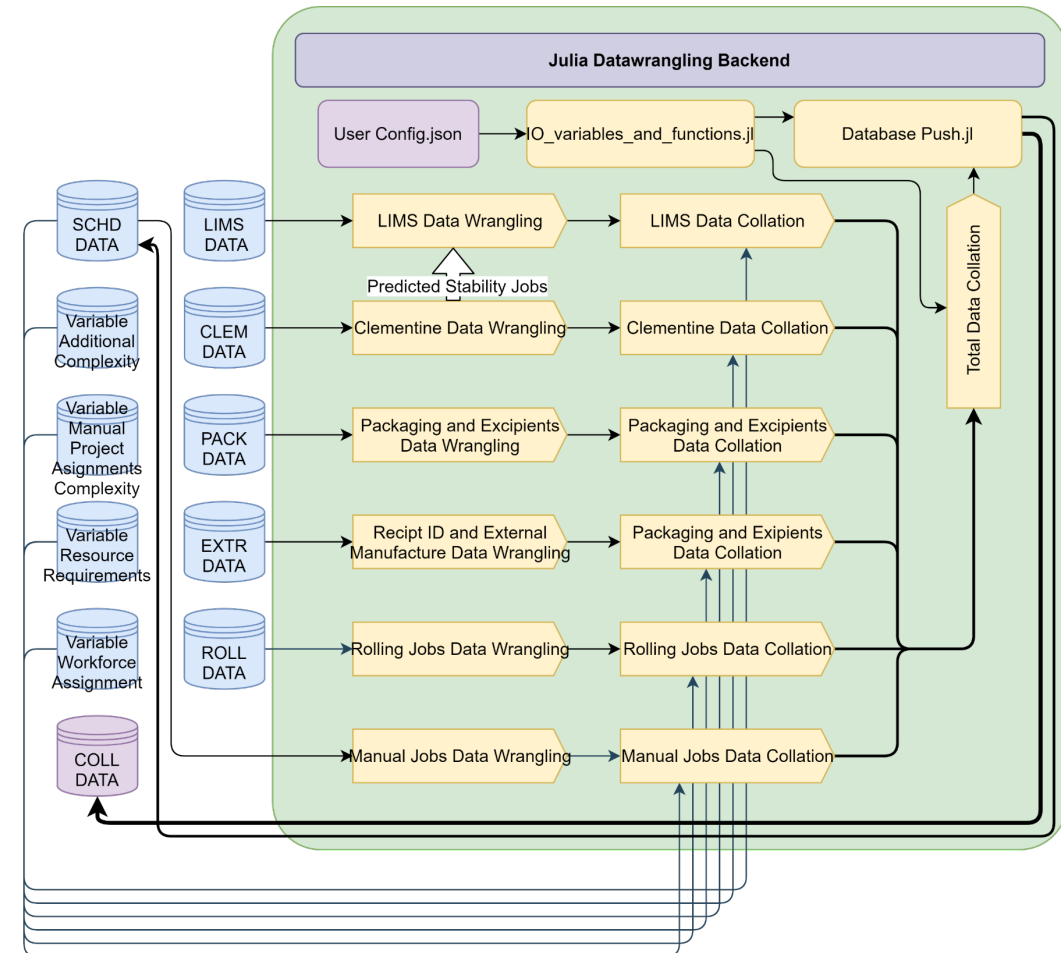
Data Table Name	Database Function
LIMS DATA	Project stability data
CLEM DATA	Project release data
PACK DATA	Project packaging, excipients data
EXTR DATA	Project external manufacturing, receipt ID data
ROLL DATA	Rolling jobs that repeats monthly
Variable Additional Complexity	User adjustable project counts economy of scales modifier
Variable Manual Project Assignments Complexity	Stores the list of all currently available projects and their individual complexity modifier
Variable Resource Requirements	User adjustable project types dependent FTE days and cycle time
Variable Workforce Assignment	Stores the current workforce (employees) FTE allocation and team assignments
COLL DATA	A full output database for Spotfire visualisation
SCHD DATA	A simplified output database for job assignment

Component 3: Julia Backend

Background Scheduled Data Processing Using Docker Containerised Julia

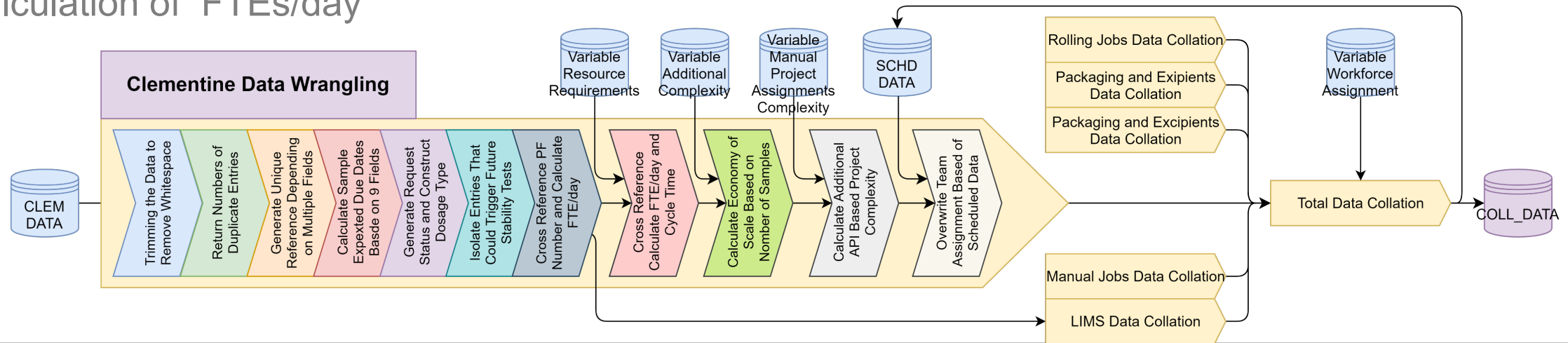
Containerised Julia Data Wrangling Backend

- Julia is a high-level programming language like Python with the speed of C. Choosing Julia for data wrangling dramatically reduced the computational time needed.
- Data from 10 databases are fetched, calculated for FTE/Days depending on a variety of variables, and populated as **COLL DATA** for data visualisation and **SCHD DATA** for GROWMAT's workload scheduler.
- Julia service is containerised along with the PostgreSQL database, allowing seamless connection between the two without the need of configuration.



Component 3: Julia Backend

Calculation of FTEs/day



Data Wrangling Pipeline (e.g. Clementine)

- Using Clementine databases as an example, raw data is fetched from **CLEM DATA**, formatted, removed duplicates, and assigned unique references. Expected sample due dates are calculated for each job depending on specific fields. Entries that might result in future stability tests are identified and sent to the process that performs LIMS data wrangling. The **FTE Days** and **Cycle Time** each type of job take is referenced from **Variable Resource Requirements**, additional economy of scale adjustment calculations are referenced and performed to the jobs' **FTEs/Day** from **Variable Additional Complexity**. **Variable Manual Project Assignments Complexity**, a PF number specific adjustment parameter is also referenced for any last-minute complexity multiplier for the **FTEs/Day** calculation. Scheduled data database **SCHD DATA** with user assigned Teams and **FTEs/day** then overwrites the automatic source. **Variable Workforce Assignment** calculates the current overall workload capacity. Wrangled data is then sent to **COLL DATA** and back into **SCHD DATA**.

Component 4-5: Ofelia Task Scheduler and Nginx Proxy Manager

Other Docker Containerised Services Essential to GROWMAT's Functionality

Component 4: Ofelia Task Scheduler

- Ofelia Task Scheduler schedules tasks to routinely run based on a timed schedule.

Name	Function	Schedule
Julia Backend	To process data	Every 2 hours
PostgreSQL Database	To back up the PostgreSQL Database	Daily with 180 days of retention
Log Trimmer	To keep the logfiles' size down	Every 3 Days

Component 5: Nginx Proxy Manager

- For the website to be accessible to Pfizer internal network, Nginx Proxy Manager points the domain name <http://gacapacity.pfizer.com/> (only accessible via Pfizer intranet) to the web server within the docker container.



NGINX
PROXY MANAGER

Component 6: GROWMAT Web service

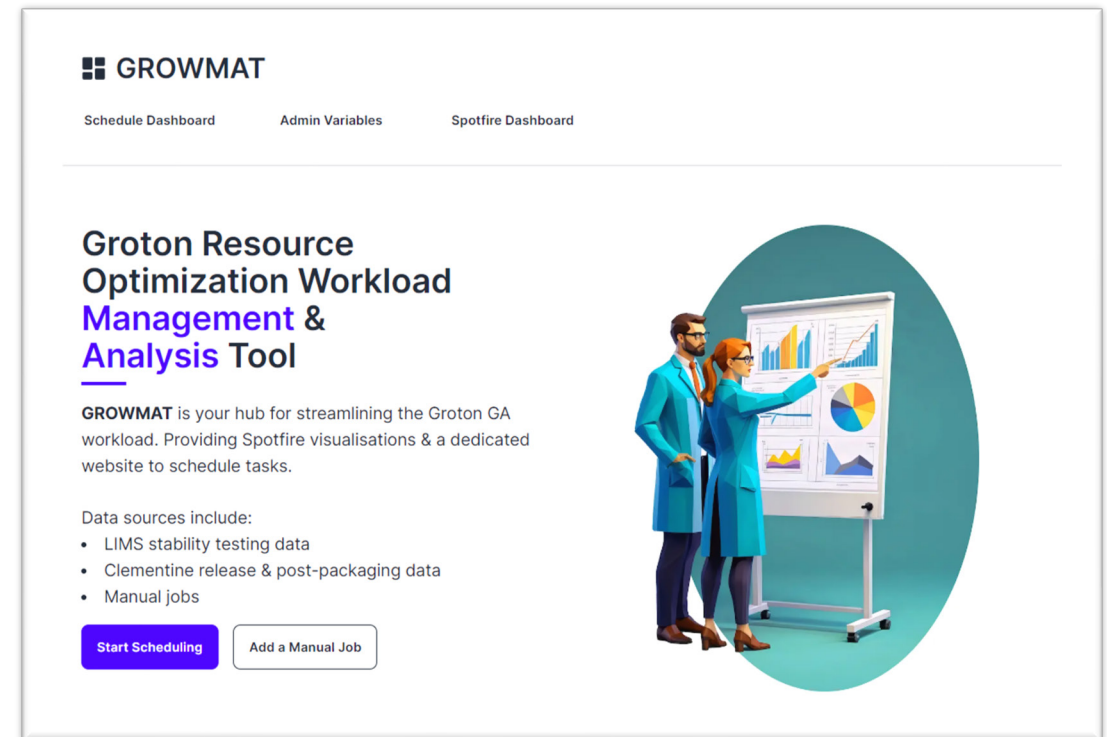
Docker Containerised Full Stack Web Server

The Technology Stack

- Next.js is a full stack framework based on React, supported by Vercel
- It runs on top of Node.js
- Provides frontend and backend services, allowing for data fetching, caching, route handlers, service actions & more

NEXT.JS

<http://gacapacity.pfizer.com/>



Component 7: GROWMAT Spotfire Visualisation

Spotfire Visualisation Independent GoCloud Server

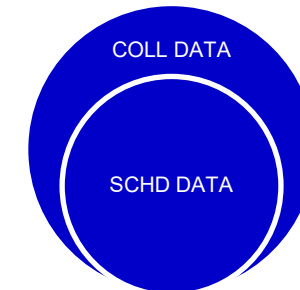
Spotfire Data Canvas View

The screenshot shows the Spotfire Data Canvas interface. At the top, there's a toolbar with icons for 'Filter rows', 'Rename', 'Replace', 'Delete', and 'Show in user interface'. Below the toolbar, a data connection icon labeled 'SCHD_DATA (PostgreSQL)' is connected to a data table icon labeled 'Part Due'. A tooltip for the data table icon says 'This is the data table used in the worksheet'. At the bottom, there's a table with columns: Job Descr..., Job Refer..., Sample Av..., Text due d..., Sample O..., Manual O..., Lifecycle, Analyst, Reviewer, Notes, is urgent, GT's record, Job Status, Number of..., FTE Days, Team, Age of Job..., Product, Mfg. The table contains 24 rows of data.

Job Descr...	Job Refer...	Sample Av...	Text due d...	Sample O...	Manual O...	Lifecycle	Analyst	Reviewer	Notes	is urgent	GT's record	Job Status	Number of...	FTE Days	Team	Age of Job...	Product	Mfg
Release (N)	1803/2024	1804/2024	1804/2024	1804/2024	Complete	Complete				No	No	Complete	1.00	8.00	LAB07	88	241	241
Release (N)	1804/2024	2205/2024	2405/2024	2405/2024	Complete	Complete				No	No	Complete	1.00	3.00	LAB09	98	241	241
Release (N)	2803/2024	2703/2024	2803/2024	2803/2024	Complete	Complete				No	No	Complete	8.00	12.00	LAB07	88	231	231
Release (N)	0304/2024	0105/2024	0305/2024	0305/2024	Complete	Complete				No	No	Complete	4.00	12.00	LAB05	98	231	231
Release (N)	0105/2024	0105/2024	0305/2024	0305/2024	Complete	Complete				No	No	Complete	1.00	7.50	LAB08	71	ms	ms
Release (N)	2802/2024	1004/2024	1204/2024	1204/2024	Complete	Complete				No	No	Complete	1.00	7.50	LAB05	108	241	241
Release (N)	1303/2024	1804/2024	1204/2024	1204/2024	Complete	Complete				No	No	Complete	1.00	7.50	LAB05	94	241	241
Release (N)	1204/2024	0805/2024	1005/2024	1005/2024	Complete	Complete				No	No	Complete	1.00	7.50	LAB05	84	241	241
Release (N)	1004/2024	2405/2024	2504/2024	2504/2024	Complete	Complete				No	No	Complete	1.00	7.50	LAB05	88	241	241
Release (N)	0803/2024	0803/2024	1005/2024	1005/2024	Complete	Complete				No	No	Complete	1.00	7.50	LAB05	88	241	241
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Release (N)	0310/2023	1511/2023	1711/2023	1711/2023	Complete	Complete				No	No	Complete	1.00	10.00	LAB08	258	231	231
Release (N)	0809/2023	1511/2023	1711/2023	1711/2023	Complete	Complete				No	No	Complete	1.00	6.00	LAB07m	281	231	231
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Release (N)	1202/2024	1804/2024	1204/2024	1204/2024	Complete	Complete				No	No	Complete	1.00	6.00	LAB07	114	241	241
Release (N)	0910/2023	1511/2023	1711/2023	1711/2023	Complete	Complete				No	No	Complete	1.00	6.00	LAB07m	250	231	231
Release (N)	2801/2024	2802/2024	2802/2024	2802/2024	Complete	Complete				No	No	Complete	5.00	12.00	LAB07	128	ms	ms
Release (N)	2604/2024	1505/2024	1505/2024	1505/2024	Complete	Complete				No	No	Complete	1.00	6.00	LAB05	98	231	231
Release (N)	0310/2023	3001/2024	0105/2024	0105/2024	Complete	Complete				No	No	Complete	4.00	12.00	LAB07	287	231	231
Release (N)	1004/2024	1505/2024	1705/2024	1705/2024	Complete	Complete				No	No	Complete	1.00	6.00	LAB05	88	241	241

Spotfire Visualisation

- The Spotfire Visualisation fetches from the GROWMAT's PostgreSQL database tables **COLL DATA** and **SCHD DATA**. The only difference between the two databases is **COLL DATA** contains 53 columns, and **SCHD DATA** contains all the same rows as **COLL DATA** but with only the necessary fields essential for job scheduling. With the two databases, the end user can build any visual or apply any custom filter without affecting the database.



Permissions Table

A Comprehensive Table Detailing the Permissions and Tasks Assigned to Each Role

Standard User Permissions

1. View, add manual jobs, and edit (not delete) existing jobs in GROWMAT Scheduler
2. View Spotfire Visualisations
3. Download Schedule for user's own analysis

Elevated User Permissions

1. All the permissions of the Standard User
2. Granted database elevated permission user account, access databases using DBeaver
3. Edit locked fields on the GROWMAT Scheduler, and delete entries under **COLL DATA** and **SCHD DATA**
4. Edit, add, and delete specific economy of scale multiplier under **Variable Additional Complexity**
5. Edit, add, and delete additional PF numbers under the **Variable Manual Project Assignments Complexity** data table
6. Edit, add, and delete project type specific FTE Days and cycle time under **Variable Resource Requirements**
7. Edit, add, and delete employee FTE availability within teams under **Variable Workforce Assignment**

Admin Permissions

1. All the permissions of the Elevated User
2. Read, write, delete, create database, grant roles options for all database items
3. Able to roll back database recovery up to 180 days
4. Able to roll back the server as disaster recovery plan
5. Able to change any code base and redeploy for any other server



Acknowledgements and Conflicts of Interest

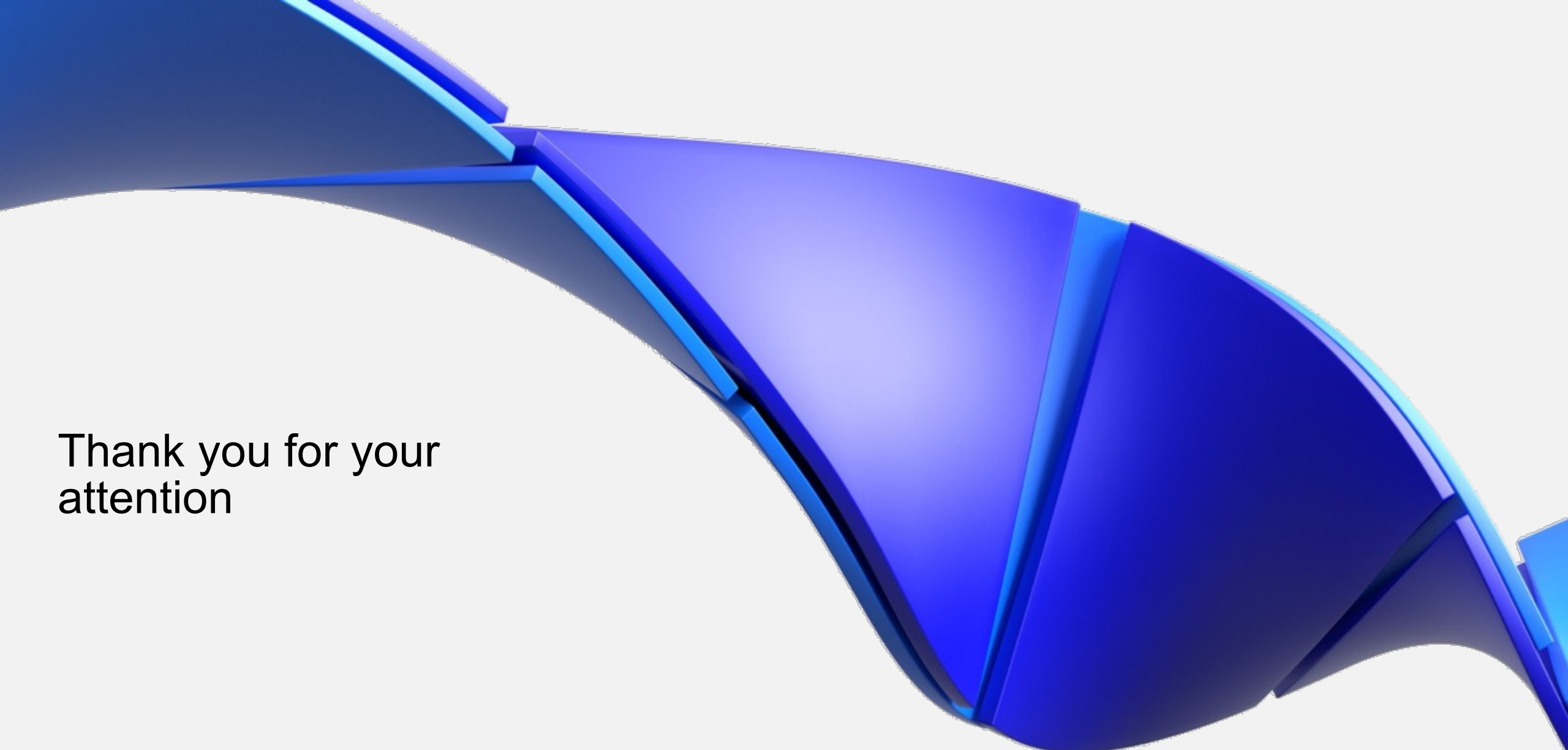
GROWMAT

Acknowledgements

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- Chang, Qing
- Glover, Darren
- Dyer, Dana Marie

Conflicts of Interest

- None

A large, abstract graphic composed of several overlapping, curved, and faceted planes in various shades of blue and purple. The shapes create a sense of depth and movement, resembling a stylized, flowing architectural structure or a molecular model. The lighting is soft, highlighting the edges and surfaces of the planes.

Thank you for your
attention